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Question

The least squares generalized inverse of an overdetermined problem is expressed as

1 $(\mathbf{G}^\top \mathbf{G})^{-1} \mathbf{G}^\top$

2 $(\mathbf{G}^\top \mathbf{G})^{-1}$

3 $\mathbf{G}^\top (\mathbf{G} \mathbf{G}^\top)^{-1}$

4 $(\mathbf{G} \mathbf{G}^\top)^{-1}$

Theoretical Solution

Let us consider the overdetermined system of equations,

$$\mathbf{G}\mathbf{x} = \mathbf{b} \quad (1)$$

Define the residual vector as,

$$\mathbf{r} = \mathbf{b} - \mathbf{G}\mathbf{x} \quad (2)$$

The least-squares problem is to minimize the squared error:

$$\mathbf{r}^\top \mathbf{r} = (\mathbf{b} - \mathbf{G}\mathbf{x})^\top (\mathbf{b} - \mathbf{G}\mathbf{x}) \quad (3)$$

$$\mathbf{r}^\top \mathbf{r} = (\mathbf{b}^\top - \mathbf{x}^\top \mathbf{G}^\top)(\mathbf{b} - \mathbf{G}\mathbf{x}) \quad (4)$$

$$\mathbf{r}^\top \mathbf{r} = \mathbf{x}^\top \mathbf{G}^\top \mathbf{G}\mathbf{x} - \mathbf{x}^\top \mathbf{G}^\top \mathbf{b} - \mathbf{b}^\top \mathbf{G}\mathbf{x} + \mathbf{b}^\top \mathbf{b} \quad (5)$$

Theoretical Solution

Now, to minimize $\mathbf{r}^\top \mathbf{r}$, define

$$f(\mathbf{x}) = \mathbf{x}^\top \mathbf{G}^\top \mathbf{G} \mathbf{x} - 2\mathbf{x}^\top \mathbf{G}^\top \mathbf{b} + \mathbf{b}^\top \mathbf{b} \quad (6)$$

Taking gradient with respect to $\mathbf{x}$:

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = 2\mathbf{G}^\top \mathbf{G} \mathbf{x} - 2\mathbf{G}^\top \mathbf{b} \quad (7)$$

Setting $\nabla_{\mathbf{x}} f(\mathbf{x}) = 0$:

$$\mathbf{G}^\top \mathbf{G} \mathbf{x} = \mathbf{G}^\top \mathbf{b} \quad (8)$$

Therefore, the least squares solution is

$$\mathbf{x} = \boxed{(\mathbf{G}^\top \mathbf{G})^{-1} \mathbf{G}^\top \mathbf{b}} \quad (9)$$

Therefore, $(\mathbf{G}^\top \mathbf{G})^{-1} \mathbf{G}^\top$ is the least squares generalized inverse of an overdetermined problem.